

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I V-RUPASHREE(192319012) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

 

Using the above constraints and BFS Search Algorithm, show step-by-step node expansion and frontier updates, and determine the optimal path and cost.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

$$25 + 25 + 23 + 12 + 10 = 95$$

CO1 - AT2

Backtracking Search - Individual Teacher Scheduling -

Problem: Assign 3 teachers (T_1, T_2, T_3) to 3 exam slots (Morning, Afternoon, Evening) subject to the given constraints.

Constraints:-

1. T_1 cannot investigate the evening slot
2. T_2 must investigate before T_3 . (Morning < Afternoon < Evening)
3. Only one teacher per slot.
4. T_3 cannot investigate the Morning slot.
5. All teachers must be assigned exactly one slot.

Backtracking Steps:-

Step	Assignment	Constraint check
1.	$T_1 \rightarrow$ morning	valid
2.	$T_2 \rightarrow$ Afternoon	T_2 before T_3 satisfied
3.	$T_3 \rightarrow$ Evening	T_3 not morning, T_2 before T_3

Final Valid Schedule

Slot	Teacher
Morning	T_1
Afternoon	T_2
Evening	T_3

Conclusion: All constraints are satisfied
Therefore, the above is the final valid investigation schedule.

2) BFS Search Algorithm - Farm Irrigation Robot

Problem:-

A farm irrigation robot operates in a 6×6 field grid from Start (S) to Goal (G). Movement is allowed up, down, left & right. obstacle cells cannot be crossed. BFS must be used to find the shortest path.

Assumed 6×6 Grid:-

	1	2	3	4	5	6
1	S		X			
2			X		X	
3	X				X	
4		X	X			
5				X		
6						G
7		X				

BFS NODE Expansion:-

Initial Queue [S]

- Expand S (1,1)
 - Add (1,2), (2,1)

Queue = [(1,2), (2,1)]

- Expand (1,2)

Add (2,2)

Queue = [(2,1), (2,2)]

Expand (2,1)

No new node

Queue = [(2,2)]

4) Expand (2,2)

Add (3,2)

Queue = [(3,2)]

5) Expand (3,2)

Add (3,3)

Queue = [(3,3)]

6) Expand (3,3)

Add (3,4)

Queue = [(3,4)]

7) Expand (3,4)

Add (2,4), (4,4)

Queue = [(2,4), (4,4)]

8) Expand (4,4)

Add (4,5)

Queue = [(2,4), (4,5)]

9) Expand (4,5)

Add (4,6), (5,5)

Queue = [(2,4), (4,6), (5,5)]

10) Expand (4,6)

Add (5,6)

Queue = [(5,5), (5,6)]

11) Expand (5,6)

Add Goal (6,6)

Optimal Path Found

$S \rightarrow (1,2) \rightarrow (2,2) \rightarrow (3,2) \rightarrow (3,3) \rightarrow (3,4) \rightarrow (4,4)$

$\rightarrow (4,5) \rightarrow (4,6) \rightarrow (5,6) \rightarrow G$

